

$$\lambda = \frac{2\pi}{k} = \frac{2\pi}{8.00 \text{ cm}^{-1}} = 2\pi \text{ cm}$$

16 A The wavelength of the electromagnetic wave is $c = \lambda f \rightarrow \lambda = \frac{3 \times 10^8}{10^{13}} = 30 \text{ } \mu\text{m}$, which is longer than the visible light. Hence, the answer is infra-red waves.

17 C ;